

Ekaterina ILIN

astrophysicist specializing in star-planet interactions

Academic positions

- 2026
→ongoing **Max-Planck Research Group Leader @ Max-Planck Institute for Solar System Research**
Magnetic Interaction of (Sub)Stellar Companion Systems
- 2023 →2026 **Postdoc @ Netherlands Institute for Radio Astronomy (ASTRON), Netherlands**
Project STORMCHASER; PI: H. K. Vedantham
- 2022 →2023 **Postdoc @ Leibniz Institute for Astrophysics Potsdam (AIP), Germany**
Funded by the German Aerospace Center (Deutsches Zentrum für Luft und Raumfahrt); PI: E. Ilin

Education

- 2018 →2022 **PhD in Astrophysics @ Leibniz Institute for Astrophysics Potsdam (AIP), Germany**
Supervisor: Prof. Dr. Katja Poppenhäger
High lights: stellar flares as probes of magnetism in stars and star-planet systems
"summa cum laude"

→ I specialize in the magnetism of small stars, its origins, long-term evolution, and interaction with the planets in their orbit.
- 2016 →2018 **Master of Science in Astrophysics @ University of Potsdam, Germany**
Supervisor: Prof. Dr. Klaus Strassmeier, mentor: Dr. Sarah J. Schmidt
Flares in clusters using K2 data
1,0 "with distinction"
- 2012 →2016 **Bachelor of Science in Physics @ Karlsruhe Institute for Technology, Germany**
Specialization in biophysics, Supervisor: Prof. Dr. Ulrich Nienhaus
Uptake of TAT-coated gold nanoclusters in live HeLa cells
1,4 "very good"

First author refereed publications

2 key publications **highlighted**

Searching for planet-induced radio signal from the young close-in planet host star HIP 67522

Ilin, E.; Bloor, S.; Callingham, J. R.; Vedantham, H. K.

2025, *Astronomy & Astrophysics*, 699, A147 doi:10.1051/0004-6361/202554684 arXiv:2507.00796

Close-in planet induces flares on its host star

Ilin, E.; Vedantham, H. K.; Poppenhäger, K.; Bloor, S.; Callingham, J. R.; Brandeker, A.; Chakraborty, H.

2025, *Nature*, **643**, 645-648 doi:10.1038/s41586-025-09236-z arXiv:2507.00791

Detected the first confident case of planet-induced flaring from time-domain observations with TESS and CHEOPS through Bayesian analysis of flare clustering near the innermost planet's transit.

The corona of a fully convective star with a near-polar flare

Ilin, E.; Poppenhäger, K.; Stelzer, B.; Dsouza D.

2024, *Astronomy & Astrophysics*, 687/A138 doi:10.1051/0004-6361/202449541 arXiv:2405.05580

Planetary Perturbers: Flaring star-planet interactions in Kepler and TESS

Ilin, E.; Poppenhäger, K.; Chebly, J.; Ilić N.; Alvarado-Gómez, Julián D.

2023, *Monthly Notices of the Royal Astronomical Society*, 523/3, pp 4326-4339 doi:10.1093/mnras/stad3398 arXiv:2311.04316

Flaring Latitudes in Ensembles of Low Mass Stars

Ilin, E.; Angus, R.; Luger, R.; Morris, B.; Jehn, F.U.

2023, *Monthly Notices of the Royal Astronomical Society*, 523/3, pp 4326-4339 doi:10.1093/mnras/stad1690 arXiv:2306.02007

Searching for star-planet interactions in AU Mic TESS observations

Ilin, E.; Poppenhäger, K.

2022, *Monthly Notices of the Royal Astronomical Society*, 513/3, pp 4579-4586 doi:10.1093/mnras/stac1232 arXiv:2204.14090

Localizing flares to understand stellar magnetic fields and space weather in exo-systems

Ilin, E.; Poppenhäger, K.; Alvarado-Gómez, J. D

2021, *Astronomische Nachrichten*, doi:10.1002/asna.202101111 arXiv:2112.09676

Giant white-light flares on fully convective stars occur at high latitudes

Ilin, E.; Poppenhäger, K.; Schmidt, S. J.; Järvinen, S. P.; Newton, E. R.; Alvarado-Gómez, J. D; Pineda, S. J.; Davenport, J. R. A.; Oshagh, M.; Ilyin, I.

2021, *Monthly Notices of the Royal Astronomical Society*, 507/2, pp 1723–1745 doi:10.1093/mnras/stab2159 arXiv:2108.01917

First determination of near-polar flaring in four rapidly rotating M dwarfs. Developed new method for localizing flare footpoints on low mass stars using optical time-domain observations and high-resolution optical spectroscopy.

AltaiPony – Flare science in Kepler, K2 and TESS light curves

Ilin, E.

2021, *Journal of Open Source Software*, 6(62), 2845 10.21105/joss.02845

Peer-reviewed open source python toolbox for statistical flare analysis in time-domain photometry.

Used in > 30 refereed publications (2025).

Flares in Open Clusters with K2. II. Pleiades, Hyades, Praesepe, Ruprecht 147, and M67

Ilin, E.; Schmidt, S. J.; Poppenhäger, K.; Davenport, J. R. A.; Kristiansen, M. H.; Omohundro, M.

2021, *Astronomy & Astrophysics*, 645, A42, 25 pp. doi:10.1051/0004-6361/202039198 arXiv:2010.05576

Flares in Open Clusters with K2. I. M45 (Pleiades), M44 (Praesepe) and M67

Ilin, E.; Schmidt, S. J.; Davenport, J. R. A.; Strassmeier, K. G.

2019, *Astronomy & Astrophysics*, 622, A133/16. doi:10.1051/0004-6361/201834400 arXiv:1812.06725

Refereed publications with significant contribution, or as supervisor

Searching for gamma-ray emission from stellar flares

Song Yuzhe; Paglione Timothy A. D., Ilin E.

2024, *Monthly Notices of the Royal Astronomical Society*, 531/3, 3215–3221 doi:10.1093/mnras/stae1347 arXiv:2405.15435

Lower than expected flare temperatures for TRAPPIST-1

Maas A. J.; Ilin E.; et al.

2022, *Astronomy & Astrophysics*, 928/2, 147. doi:10.1051/0004-6361/20224386 arXiv:2210.11103

Refereed publications with minor contributions

Occurrence rate of stellar Type II radio bursts from a 100 star-year search for coronal mass ejections

Konijn, D.; ...; Ilin, E.; ...; et al.

2025, *Astronomy & Astrophysics*, 703, id.A198, 13 pp. doi:10.1051/0004-6361/202554317 arXiv:2511.09296

Polarized, variable radio emission from the scallop-shell binary system DG CVn

Kaur, S.; ...; Ilin, E.; ...; et al.

2025, *Astronomy & Astrophysics*, 701, A69 doi:10.1051/0004-6361/202555222 arXiv:2507.09366

Hints of auroral and magnetospheric polarized radio emission from the scallop-shell star 2MASS J05082729-2101444

Kaur, S.; ...; Ilin, E.; ...; et al.

2024, *Astronomy & Astrophysics Letters*, 691/L17 9 pp. doi:10.1051/0004-6361/202452037 arXiv:2410.22449

Seaweed as a resilient food solution in nuclear winter

Jehn F. U.; Jasmine Dingal, F.; Mill, A.; Ilin, E.; Harrison, C.; Roleda, M. Y. ; Denkenberger, D.

2024, *Earth's Future*, 12/1 doi:10.1029/2023EF003710

Three young planets around the K-dwarf K2-198: high-energy environment, evaporation history, and expected future

Ketzer, L.; Poppenhaeger, K.; Baratella, M.; Ilin, E.

2024, *Monthly Notices of the Royal Astronomical Society*, 527/1, pp. 374–385 doi:10.1093/mnras/stad3197 arXiv:2311.06897

Focus of the IPCC Assessment Reports has shifted to lower temperatures

Jehn F. U.; Kemp L.; Ilin E.; Funk C.; Wang J. R.; Breuer L.

2022, *Earth's Future*, 5/10 doi:10.1029/2022EF002876

Simulating the Space Weather in the AU Mic System: Stellar Winds and Extreme Coronal Mass Ejections

Alvarado-Gómez, J. D; Cohen O.; Drake J. J.; Fraschetti F.; Poppenhäger, K.; Garraffo C.; Chebly J.; Ilin, E.; Harbach

L.; Kochukhov O.

2022, The Astrophysical Journal, 928/2, 147. doi:10.3847/1538-4357/ac54b8 arXiv:2202.07949

Teaching and supervision

total supervised: 3 Master level, 2 Bachelor level

2025 → 2026	Supervising students @ Kapteyn Institute Supervised the BSc project of Andrei Danila, and the MSc project of Casper Hindriks.	Groningen, NL
2024	Supervising summer student @ ASTRON Supervising a 10-week summer student project with undergraduate student Necdet Canim (Pomona College, CA, USA).	Dwingeloo, NL
2024	Invited lecturer at summer school @ ICE-CSIC Institute of Space Sciences On the role of flares in star-planet interactions	Barcelona, Spain
2023	Lecturer at summer school @ German National Scholarship Foundation Teaching undergraduates about exoplanets in an interdisciplinary context	Vienna, Austria
2023	Invited lecturer @ Tag der Schulastronomie Sachsen Lecture about exoplanets for high school physics teachers in Saxonia.	Meißen, Germany
2022 → 2025	Mentoring graduate student Cinta Vidante @ University of Potsdam () Graduated with distinction; publication in prep.	Potsdam, Germany
2020 → 2022	Mentoring graduate student Aaron Maas @ University of Heidelberg Maas, Ilin et al. (2022). Now PhD student in astrobiology at CSIC-ICE.	remote
2020	Teaching assistant @ University of Potsdam Galaxies and Cosmology undergraduate course	Potsdam, Germany
2016 → 2018	Teaching assistant @ University of Potsdam Introduction to mathematical methods in physics	Potsdam, Germany

External funding

total ca. 3,000,000€

2026 → ongoing	PI of Max-Planck Research Group Junior research group funding, 2,500,000€	Max-Planck Society
2022 → 2023	PI of DLR Grant "XrayLoops" Two years of postdoctoral funding, ca. 140,000€	Deutsches Zentrum für Luft- und Raumfahrt (DLR)
2021 → 2022	Fulbright Scholarship @ American Museum of Natural History Awarded to doctoral candidates. Includes a stipend, travel grant, and academic support throughout a six month research stay in the USA, ca. 12,000€	Fulbright Commission
2020 → 2022	Doctoral Scholarship Includes a stipend, travel grants, and academic support throughout the doctoral studies, ca. 55,000€	German Academic Scholarship Foundation
2012 → 2018	Full Scholarship Includes a stipend, travel grants, and academic support throughout the undergraduate and graduate studies, ca. 60,000€	German Academic Scholarship Foundation

Awards and prizes

2024	Stargazer Award 2024 5k€ awarded to outstanding PhD theses in astronomy in German speaking countries.
2024	Publication Prize @ Leibniz-Kolleg Potsdam 2.5k€ awarded to outstanding PhD theses in the natural science in Berlin and Brandenburg.
2019	Graduation Award @ Physikalische Gesellschaft zu Berlin, Berlin, Germany 1.5k€ awarded to excellent master theses in physics earned at universities in Berlin and Potsdam.

2014 **Erasmus+ Scholarship @ Universitet i Oslo, Norway**
 Awarded funding for six months of undergraduate studies. Focus: space physics, computational physics

Observing grants as Principal Investigator

2026 **La Silla 3.6m telescope / HARPSpol, 28h**
 PI: **E. Ilin**; Co-Is: Stefano Bellotti, Aline Vidotto, Alexis Lavail, Pascal Petit

2025 **XMM-Newton Large Program, 612ks Priority A**
 PI: **E. Ilin**; Co-PIs: K. Poppenhäger, R. Osten, A. Strugarek; Co-Is: Nikoleta Ilić, H. K. Vedantham

2025 **Submillimeter Array, 10 hours DDT**
 PI: **E. Ilin**, Co-I: J. Lovell

2025 **CHEOPS, 270 orbits, Priority 1**
 PI: **E. Ilin**

2024 **VLT/UVES, 6.4 hours**
 PI: **E. Ilin**, Co-Is: H. K. Vedantham, S. Järvinen

2024 **Australian Telescope Compact Array, 144 hours**
 PI: **E. Ilin**, Co-Is: S. Bloor, J. Callingham

2023 **CHEOPS, 128 orbits, Priority 1**
 PI: **E. Ilin**, Co-Is: K. Poppenhäger

2021 **XMM-Newton, 36ks priority A**
 PI: **E. Ilin**, Co-Is: K. Poppenhäger, B. Stelzer

Observing grants as Co-Investigator

2025 **GMRT, 32 hours**
 PI: S. Kaur, Co-Is: D. Vigano, **E. Ilin** and 14 others

2020 **SALT HRS, 2.8 hours DDT**
 PI: K. Poppenhäger, Co-Is: **E. Ilin**, S. J. Schmidt

Invited talks at conferences

2026 **COSPAR** Florence, Italy
 Magnetic star-planet interactions: Teenage planets under the influence of their host stars

2025 **EAS Annual Meeting** Cork, Ireland
 Review: Magnetic star-planet interactions

2024 **NOVA Symposium** Den Haag, NL
 Magnetic winds and plasma seas: Interactions in star-planet systems

2023 **All-hands-on-deck meeting – German Exoplanet Diversity Group** Munich, Germany
 How close-in planets manipulate their host stars

2023 **From the Heliosphere to Astrospheres - Lessons for Exoplanets and their Habitability** Bad Honnef, Germany
 Magnetic star-planet interactions

Invited colloquia and seminar talks (selection)

2025	St. Mary's University Magnetic star-planet interactions	Halifax, CA
2025	University of Geneva Magnetic star-planet interactions: Teenage planets under the influence of their host stars	Geneva, CH
2025	Leiden Observatory Self-roast: Close-in planet induces flares on its host star	Leiden, NL
2025	Vienna University Channeling sparks through interplanetary space: Magnetic star-planet interaction	Vienna, AT
2022	University of Tübingen Flares on low mass stars: from detection to characterization, and beyond	Tübingen, GER
2022	American Museum of Natural History Flare observations: probing the magnetism of low-mass stars and star-planet systems	New York City, USA
2021	Harvard Smithsonian Center for Astrophysics Exoplanet Pizza Lunch Magnetic worlds: What flares tell us about small stars and their planets	virtual

Contributed talks (selection)

2026	CoolStars 23 Self-roast: Close-in planet induces flares on its host star	Tokyo, Japan
2026	Exoplanets 6 Self-roast: Close-in planet induces flares on its host star	Porto, Portugal
2024	CoolStars 22 Hot Jupiters and stellar eruptions: Planet-induced flares in young exoplanetary systems	San Diego, CA, USA
2024	BCool A fully convective rapid rotator with a high latitude flare and a dim corona	ASTRON, NL
2024	ESA's Holland-Area Exoplanet Science Meeting Hot Jupiters and stellar eruptions: planet-induced flares in young exoplanetary systems	Nordwijk, NL
2023	European Astronomical Society Meeting Planet-induced flares in Kepler and TESS observations	Krakow, Poland
2023	The X-ray Universe High-latitude flares on rapidly rotating M dwarfs - a special corona?	Athens, Greece
2019	Planet-Star Connections in the Era of TESS and Gaia Stellar magnetic evolution: flaring activity in K2 open clusters	Santa Barbara, CA, USA
2018	Dwarf Stars and Clusters with K2 In search of the flaring-age relation.	Boston, MA, USA

Academic service, outreach and civic involvement (selection)

2025→ongoing	Postdoc representative @ ASTRON Organizing social events and training opportunities for ASTRON early career researchers.
2025	Podcast 50-min podcast on exoplanets and the search for life for the German Bild der Wissenschaft magazine.
2025→ongoing	Lorentz Center Workshop on star-planet interactions Initiator and chair of the SOC for a 5-day workshop on star-planet interactions at the Lorentz Center in Leiden, NL.
2025	LOC @ Beyond Earth (ASTRON) Conference with 70 participants. Main tasks: Speaker liaison.
2021 →ongoing	Collaboration with various popular science magazines and blogs PM Magazin, Newsweek, New Scientist, Leibniz Magazin, Ulrike Boehm's Women in Research blog
2021 →ongoing	Referee @ various astronomical journals Nature Astronomy, Astronomy & Astrophysics, AAS Journals, Astronomy and Computing, Journal for Open Source Software, Astronomische Nachrichten
2024	Weekend workshop @ Sternwarte Berlin Planned a weekend long workshop on exoplanets for a small group of interested undergraduate students of all fields.
2023	Lecturer @ Vienna Summer School of the German Academic Scholarship Foundation Week-long interdisciplinary summer school for 20 undergraduate scholars focused on exoplanets, and the relations between exoplanet research, ethics and culture.
2021 →2022	Student representative @ Internal Scientific Committee at AIP Social events, well-being of students at the institute, on-boarding of new students.
2021	Public talks @ YouTube channel "Urknall, Weltall und das Leben" 30-min YouTube video and follow-up 12-min YouTube video (in German), ≈ 40k views total.

Training and internships

2021-2022	Career Mentoring Training in non-violent communication, conflict resolution, diversity awareness, and adaptive leadership.	Cusanuswerk
2019	3rd Advanced School on Exoplanetary Demographics Lectures on exoplanet formation, detection, demographics, star-planet interactions.	Vietri Sul Mare, Italy
2018	Visiting Scientist @ NASA Ames Research Center Awarded funding for an internship at the Kepler mission headquarters.	Mountain View, CA, USA

languages Russian, German (bilingual), English (C2), Dutch (B1), French (B1)

skills Python, Fortran, Unix, LaTeX, Git(Hub), CASA, XMM-Newton SAS, XSPEC

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